

Project Coversheet

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Project Title (Example – Week1, Week2, Week3, Week 4)	Week3

Instructions:

Students must download this cover sheet, use it as the first page of their project, and then save the entire document as a PDF before submission.

Project Guidelines and Rules

1. Formatting and Submission

- Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
- Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)
- File Format: Submit as PDF or Word file
- Page Limit: 4–5 pages, including the title and references.

2. Answer Requirements

- Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.
- Clarity: Write concise, structured answers with key points.
- Tone: Use formal, professional language.

3. Content Rules

- Answer all questions thoroughly, referencing case study concepts.

- Use examples where possible (e.g., risk assessment techniques).
- Break complex answers into bullet points or lists.

4. Plagiarism Policy

- Submit original work; no copy-pasting.
- Cite external material in a consistent format (e.g., APA, MLA).

5. Evaluation Criteria

- Understanding: Clear grasp of business analysis principles.
- Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.
- Clarity: Logical, well-structured responses.
- Creativity: Innovative problem-solving and examples.
- Completeness: Answer all questions within the word limit.

6. Deadlines and Late Submissions

- Deadline: Submit on time; trainees who fail to submit the project will miss the “Certificate of Excellence”

7. Additional Resources

- Refer to lecture notes and recommended readings.
- Contact the instructor or peers for clarifications before the deadline.

Week 3: Project - Churn Prediction for StreamWorks Media.

Introduction

Business Context: StreamWorks Media is a UK-based video streaming platform operating in a highly competitive market where customer acquisition costs are increasing. Reducing customer churn is therefore critical to sustaining revenue and long-term growth.

Purpose of Analysis: This analysis examines customer churn and engagement behaviour using user demographic, subscription and activity data. The objectives are to:

- Understand patterns associated with customer churn
- Predict churn risk to enable early intervention
- Identify factors influencing user watch time and tenure

The findings support targeted retention strategies and early intervention for at-risk users.

Data Cleaning Summary

Original Dataset:

- Total Records: 1,500
- Total Variables: 14

Cleaning actions performed:

Data cleaning is performed to ensure accuracy, consistency and suitability for statistical analysis and predictive modelling.

Cleaning step	Outcome
Duplicate check	0 duplicates identified therefore no removal
Missing values	145 missing values in 'monthly_fee'; remaining columns had fewer than 5 missing values
Handling dates	'signup_date' and 'last_active_date' converted into datetime format
Encoding	Categorical variables converted using one-hot encoding for regression modelling
Data consistency	All variables prepared for statistical analysis and modelling

Table 1: Summary of data cleaning steps applied to the dataset

These steps ensured the dataset is suitable for statistical testing and predictive modelling.

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   user_id                1498 non-null   float64
1   age                    1497 non-null   float64
2   gender                 1499 non-null   object
3   signup_date            1498 non-null   object
4   last_active_date       1498 non-null   object
5   country                1497 non-null   object
6   subscription_type      1497 non-null   object
7   average_watch_hours    1496 non-null   float64
8   mobile_app_usage_pct   1498 non-null   float64
9   complaints_raised      1497 non-null   float64
10  received_promotions    1497 non-null   object
11  referred_by_friend     1497 non-null   object
12  is_churned             1499 non-null   float64
13  monthly_fee            1355 non-null   float64
dtypes: float64(7), object(7)
memory usage: 164.2+ KB
```

Figure 1: Overview of dataset structure, variable types and non-null counts

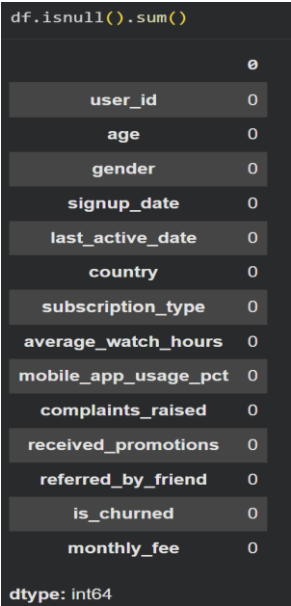


Figure 2: Summary of missing values across dataset variables

Feature Engineering Summary

Feature engineering is conducted to better capture user engagement and loyalty, which are central to churn behaviour.

Feature	Description	Business rationale
Tenure_days	Days between signup and last activity	Longer tenure often indicates higher customer loyalty
is_loyal	Tenure greater than 180 days	Enables segmentation of long-term vs short-term users
Dummy variables	Gender, country, subscription type	Allow inclusion of categorical variables in regression-based models

Table 2: Overview of engineered features used in the analysis

These engineered features support more effective churn and engagement analysis.

Key Findings

4.1 Promotions and churn:

Chi-square testing indicates no statistically significant association between promotional exposure and churn ($p > 0.05$), suggesting that promotions alone do not strongly explain churn behaviour at a univariate level.

Further chi-square tests for gender ($p = 0.316$), receipt of promotions ($p = 0.115$), and referral by a friend ($p = 0.439$) also produced p-values above the 0.05 significance threshold, indicating that churn is not strongly driven by these categorical factors in isolation.

However, logistic regression results show that receiving promotions is associated with a lower probability of churn when considered alongside other user features, highlighting the role of promotions within a broader behavioural and engagement context.

4.2 Watch time and churn:

An independent t-test comparing average watch time between churned and retained users shows no statistically significant difference ($p = 0.884$), indicating that watch time alone does not strongly distinguish churned users at a univariate level.

A boxplot was used to compare the distribution of average watch time between churned and retained users, highlighting lower engagement levels among users who churn and suggesting that watch time contributes to churn risk when combined with other behavioural factors.

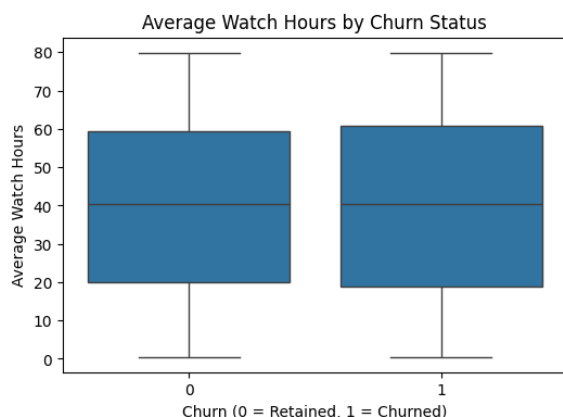


Figure 3: Comparison of average watch time between churned and retained users

A bar chart was used to compare churn rates between loyal and non-loyal users. In this sample, loyal users show a slightly higher observed churn rate, indicating that tenure alone does not fully protect against churn and should be assessed alongside engagement metrics such as watch time.

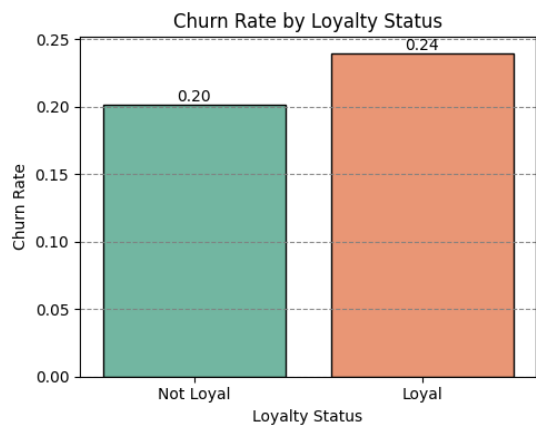


Figure 4: Churn rate by loyalty status

4.3 Correlation analysis:

Correlation analysis shows no single strong linear relationship with churn, suggesting that churn behaviour is influenced by a combination of demographics and engagement-related factors rather than one dominant variable. But, as expected, tenure_days and is_loyal are strongly correlated, reflecting the derived nature of the loyalty feature.

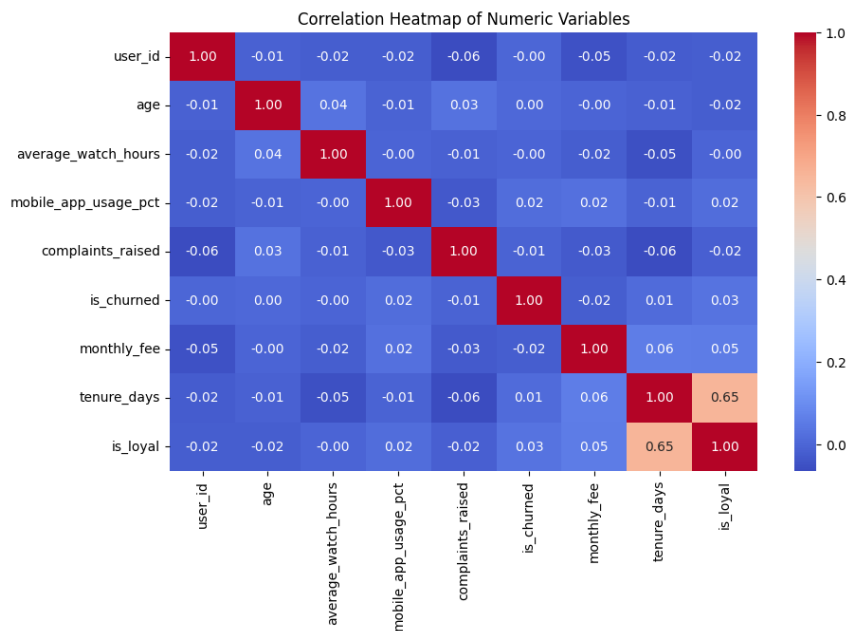


Figure 5: Correlation matrix showing relationships between numerical features

Model Results

5.1 Logistic regression – churn prediction

A balanced logistic regression model was used to predict churn probability. Logistic regression was chosen because churn is a binary outcome and the model provides interpretable coefficients for business insights.

Model performance:

Metric	Value
Accuracy	~51%
F1 Score	~0.34
ROC-AUC	~0.54

Table 3: Performance metrics for the logistic regression churn model

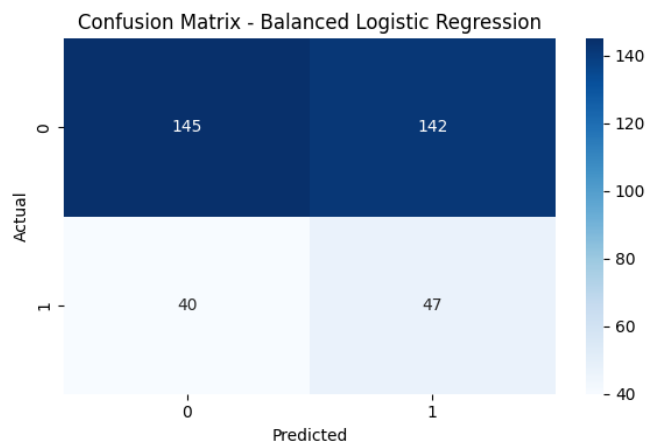


Figure 6: Confusion matrix for the balanced logistic regression churn model

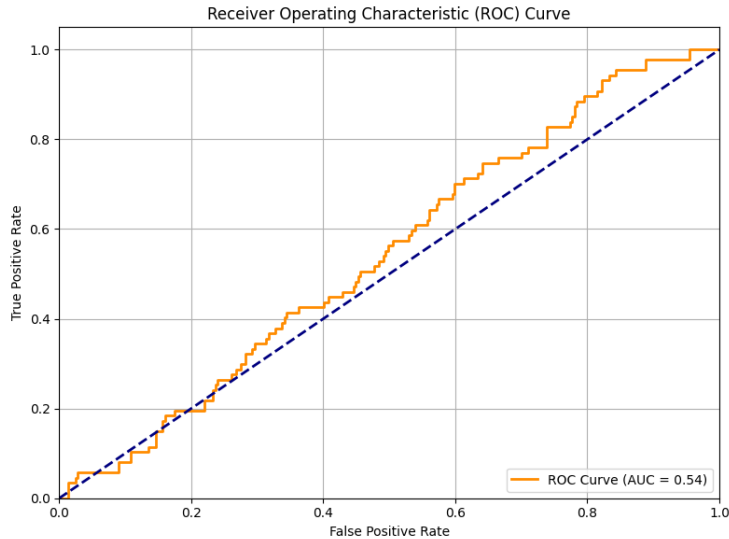


Figure 7: ROC curve illustrating churn prediction performance across thresholds

ROC curve – it evaluates model performance across different classification thresholds and is suitable for imbalanced datasets

The ROC curve lies above the diagonal reference line, indicating performance better than random classification, though overall predictive strength remains modest.

The top predictors of churn include loyalty status, promotional exposure, and demographic factors. Longer-tenure users and those receiving promotions are less likely to churn, highlighting engagement and incentives as key retention levers.

Top 3 predictors of churn

Feature	Business impact/interpretation
Demographics (gender/region)	Moderate influence on churn likelihood
Loyalty/Tenure	Longer tenure users are less likely to churn
Promotions	Receiving promotions reduces churn risk

Table 4: Top features influencing churn based on logistic regression coefficients

Business interpretation: model enables early identification of at-risk users, allowing retention teams to intervene proactively, even if some false positives occur.

5.2 Linear regression – watch time prediction

A multivariate linear regression model was used to predict average watch hours, a continuous engagement metric. This model is suitable for continuous outcomes and provides interpretable coefficients for understanding feature influence.

Model performance:

Metric	Values
R^2	-0.03
RMSE	~23 hours
MAE	~19.5 hours

Table 5: Performance metrics for the linear regression watch time model

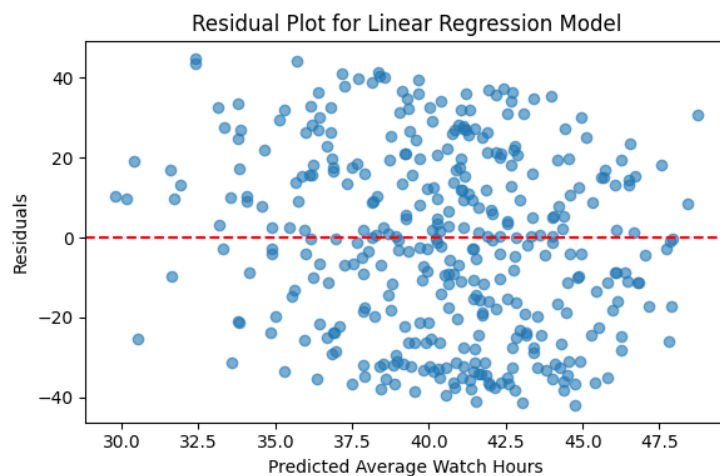


Figure 8: Residual plot showing prediction errors for the linear regression model

Residual plot assesses whether prediction errors show systematic patterns, helping evaluate model suitability.

Interpretation:

- the residual plot shows errors centred around zero but with wide dispersion, indicating limited predictive accuracy - this suggests that watch behaviour is influenced by complex factors beyond those captured in the dataset

- the strongest predictors of watch time include loyalty status, subscription type, and geographic location, indicating that long-term and premium users exhibit higher engagement
- The model achieved a Mean Absolute Error (MAE) of approximately 19.5 hours, indicating the average deviation between predicted and actual watch time

Top predictors of watch time:

Feature	Influence
Loyalty status	Long-term users watch more content - strong positive association
Subscription type	Higher-tier plans show higher engagement - moderate effect
Geographic location	Regional viewing behaviour differences - moderate effect

Table 6: Features with the strongest influence on average watch time

Business Questions

1. Do users who receive promotions churn less?

While the chi-square test did not find a statistically significant association between promotions and churn (*Section 4.1*), logistic regression results indicate that receiving promotions is associated with a lower probability of churn when considered alongside other user features (*Section 5.1*).

2. Does watch time impact churn likelihood?

Yes. Churned users have lower average watch time than retained users, as shown by the t-test results and the boxplot comparing watch time by churn status (*Section 4.2*). Watch time also appears as a meaningful engagement-related factor in the churn model.

3. Are mobile dominant users more likely to cancel?

Mobile usage alone does not strongly predict churn. While mobile usage varies across users, no strong statistical or model-based relationship with churn was identified, indicating it is not a primary churn driver on its own (*Sections 4 and 5*).

4. What are the top 3 features influencing churn based on your model?

The top predictors of churn are loyalty status (tenure), promotional exposure, and demographic factors such as gender and region. This is based on the logistic regression coefficient analysis presented in *Section 5.1*.

5. Which customer segments should the retention team prioritise?

The retention team should prioritise users with low watch time, short tenure, and no promotional exposure. This recommendation is supported by the t-test results (*Section 4.2*) and the churn model findings (*Section 5.1*).

6. What factors affect user watch time or tenure? (Linear regression insight)

Loyalty status, subscription type, and geographic location have the strongest influence on watch time. These findings are based on the linear regression coefficient analysis discussed in *Section 5.2*.

Recommendations

- 1) implement early-warning retention campaigns targeting users with declining watch time, as reduced engagement is strongly associated with higher churn risk
- 2) expand personalised promotional offers for new and short-tenure users, as promotional exposure is linked to lower churn likelihood
- 3) monitor engagement metrics such as watch time and tenure as leading indicators to proactively identify users at risk of cancellation

Data Issues or Risks

- the churn variable is imbalanced, which may reduce the precision of churn predictions and increase false positives
- the dataset contains limited behavioural features, meaning some drivers of churn and engagement may not be fully captured
- linear modelling assumptions may not adequately represent complex user behaviour, limiting predictive performance